Lesson 12.2 - The Chi Square Distribution Test of a Single Standard Deviation

> **Video:** Chi Square Distribution Test of a Single Variance or Standard Deviation by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The big picture: testing spread instead of center

Every test so far has been about a **center**: a mean or a proportion. Sometimes the question is about **consistency**:

- Does a machine fill bags more evenly after a repair?
- Are this professor's exam scores less spread out than the department's?
- Is a new investment less risky (less variable)?

These questions make a claim about the population **standard deviation** σ (or the variance σ^2), so we need a test for spread.

1. Compare the sample's spread with the claim. The natural measure is the ratio

$$\frac{s^2}{\sigma_0^2}$$

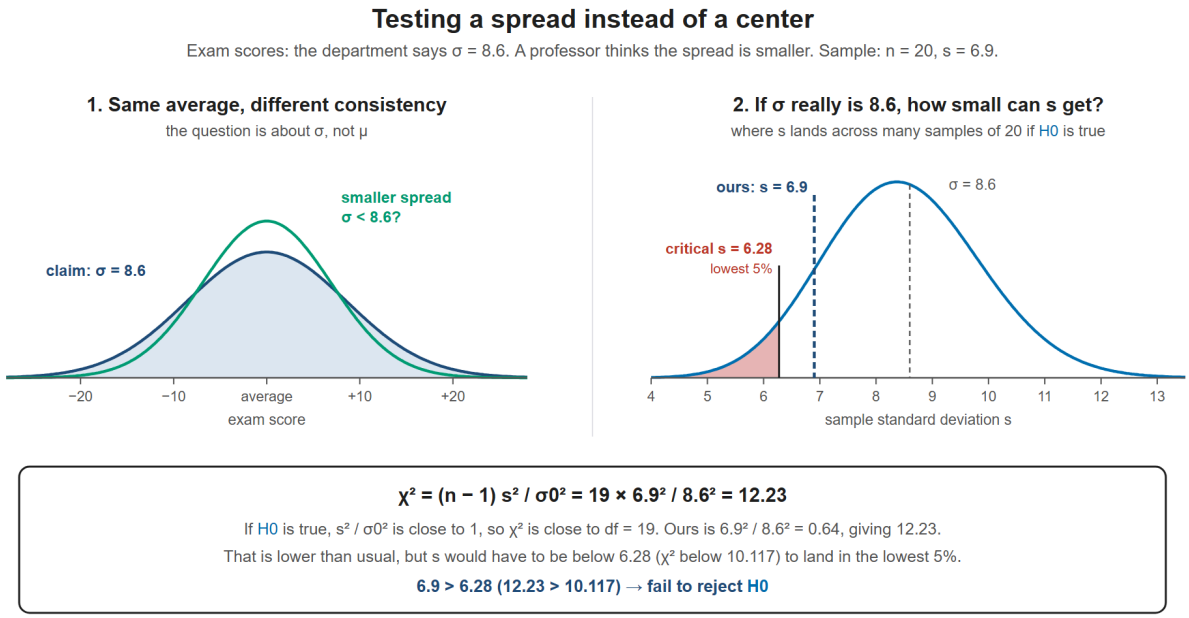
If H_0 is true, this ratio should be close to 1. A ratio well below 1 suggests a smaller spread, and well above 1 a bigger one. In the example, $6.9^2/8.6^2 = 0.64$. Is that far enough below 1?

2. Why multiply by $(n-1)$? The sample variance is $s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$, so $(n-1)s^2 = \sum(x-\bar{x})^2$. Then:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \sum\left(\frac{x-\bar{x}}{\sigma_0}\right)^2$$

That is a **sum of squared z-scores**, just like in Lesson 12.1. It follows the χ^2 distribution, provided the population is roughly normal.

3. Why df $= n-1$? The deviations $x-\bar{x}$ always add up to 0. Once you know $n-1$ of them, the last one is forced. This is the same reason that df $= k-1$ in Lesson 12.1. When H_0 is true, χ^2 averages about df. Here that's 19, and we got 12.23: lower than usual, but is it low enough?



Even when σ really is 8.6, samples of 20 give a range of s values: about 90% fall between 6.3 and 10.8. To land in the lowest 5%, s would have to be below 6.28 (that is, χ^2 below 10.117). Our $s = 6.9$ isn't low enough.

4. The curve is not symmetric. χ^2 can't be negative and is skewed right, so the left and right critical values are **not** mirror images like ± 1.96. χ^2 tables list **right-tail** areas:

- **Right-tailed test:** look up α.
- **Left-tailed test:** look up $1 - \alpha$ (here, 0.95).
- **Two-tailed test:** look up both $1 - \alpha/2$ and $\alpha/2$.

5. Variance or standard deviation, it's the same test. A claim about σ is a claim about σ^2. Square the claimed SD: $\sigma_0 = 8.6$ becomes $\sigma_0^2 = 73.96$.

The 5 steps are the same as always. Only step 4 uses a new formula:

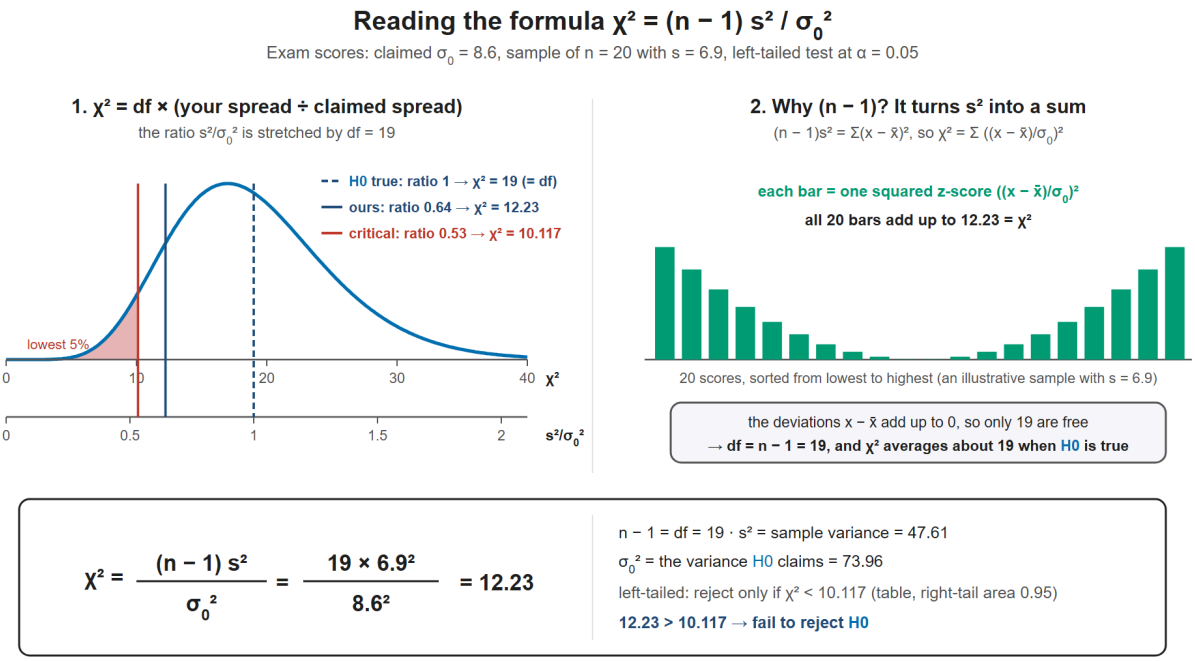
1. Write H_0 ($\sigma^2 = \sigma_0^2$) and H_a ($<$, $>$ or $\neq$).
2. Choose α.
3. Find the χ^2 critical value(s) with df $= n - 1$, using right-tail areas.
4. **Compute** $\chi^2 = (n-1)s^2/\sigma_0^2$. ← this lesson
5. In the rejection region → reject H_0. Otherwise → fail to reject H_0.

Test statistic

To test a claim about a population **variance** σ^2 (or standard deviation σ):

$$\chi^2 = \frac{(n-1)\,s^2}{\sigma_0^2}, \qquad \text{df} = n - 1$$

- s^2 = sample variance
- σ_0^2 = the variance claimed in H_0



Two ways to read the formula. Left: $\chi^2 = \text{df} \times \dfrac{s^2}{\sigma_0^2}$, the spread ratio stretched by df. A ratio of 1 (what H_0 expects) becomes χ^2 = df = 19. Our ratio of 0.64 becomes 12.23, which is not below the critical value 10.117 (ratio 0.53). Right: multiplying by $n-1$ turns s^2 back into a sum of squared deviations, so χ^2 is a sum of squared z-scores. That's why it follows the χ^2 distribution, with df $= n-1$.

H_a	Test	Rejection region
$\sigma^2 < \sigma_0^2$	Left-tailed	$\chi^2 <$ critical value, with area α on the left (look up right-tail area $1-\alpha$)
$\sigma^2 > \sigma_0^2$	Right-tailed	$\chi^2 >$ critical value, with right-tail area α
$\sigma^2 \neq \sigma_0^2$	Two-tailed	both tails, $\alpha/2$ each

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Example - is the spread of exam scores smaller?

The math department's exam scores have $\sigma = 8.6$. A professor thinks it is **less**. A sample of 20 scores has $s = 6.9$. Test at $\alpha = 0.05$.

Known values: $\sigma_0^2 = 8.6^2,\ \ s^2 = 6.9^2,\ \ n = 20,\ \ \text{df} = n - 1 = 19$

Hypotheses:

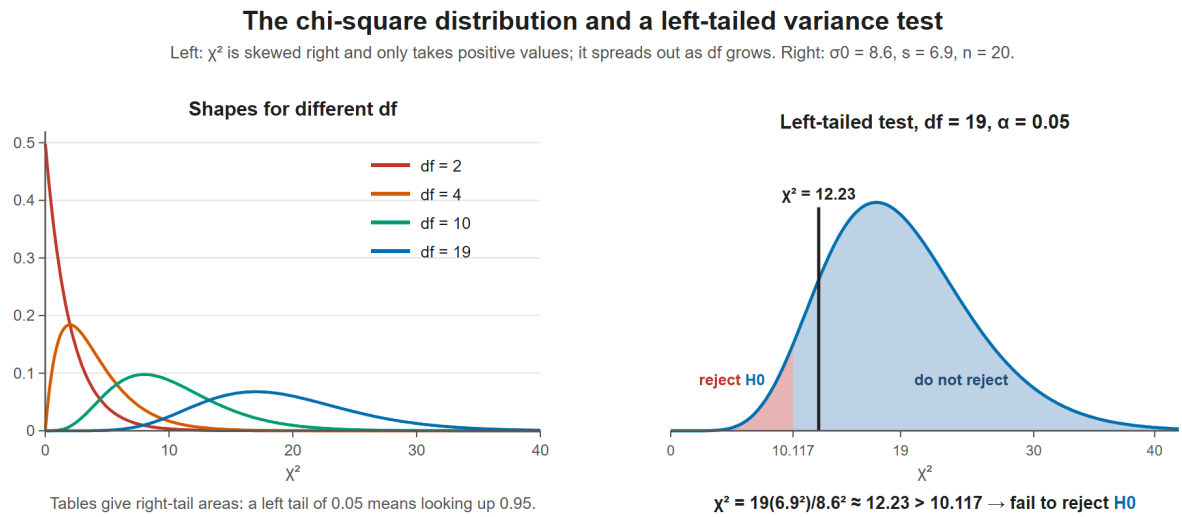
$$H_0 : \sigma^2 = 8.6^2 \qquad H_a : \sigma^2 < 8.6^2 \quad \text{(left-tailed)}$$

Critical value: the left-tail area is 0.05, so the right-tail area is 0.95. With df = 19 the table gives $\chi^2 = 10.117$. Reject H_0 only if the calculated χ^2 is **less than 10.117**.

Test statistic:

$$\chi^2 = \frac{(n-1)\,s^2}{\sigma_0^2} = \frac{19 \times 6.9^2}{8.6^2} = \frac{19 \times 47.61}{73.96} \approx 12.23$$

$12.23 > 10.117$, which is in the do-not-reject region. **Fail to reject H_0.** At the 5% level the sample does not provide enough evidence that the standard deviation is less than 8.6.



Left: χ^2 is never negative and is skewed to the right; with more degrees of freedom it moves right and spreads out. Right: for the left-tailed variance test the rejection region is the left tail below 10.117, and 12.23 does not reach it.

Key takeaways

- Testing a variance or SD: $\chi^2 = (n-1)s^2/\sigma_0^2$ with df $= n - 1$.
- If H_0 is true, $s^2/\sigma_0^2 \approx 1$, so χ^2 is near df. A χ^2 far below df suggests less spread, and far above suggests more.
- χ^2 tables list **right-tail areas**. For a left-tailed test, look up $1 - \alpha$.
- Left-tailed: reject only when χ^2 is **below** the critical value.